# Andrea Grioni, PhD


**Associate Director, Global Clinical Development · Novartis**

Basel, Switzerland · Personal website: andreagrioni.github.io · linkedin.com/in/agrioni · Google Scholar


## PROFILE

Clinical development professional working where clinical trials meet data science and AI. Building machine-learning models for biomedical data since 2014; at Novartis since 2020, moving from exploratory biomarker analysis in Phase I–III trials to the global team of a Phase III cardiovascular outcome trial. Product owner of an internal AI initiative in clinical development. Focused on making trials faster and more traceable without losing the clinical judgement they depend on.

## EXPERIENCE

**Associate Director, Global Clinical Development** 2025–present
Novartis · Basel, CH

- Global clinical trial team member for VICTORION-2 PREVENT, an international Phase III cardiovascular outcome trial of inclisiran.
- Align data management, biostatistics, operations and country teams on data capture, the statistical analysis plan and protocol execution; resolve site questions on how the protocol applies in practice.
- Work with European national leaders and country teams on patient retention and endpoint collection.
- Product owner of an internal AI initiative in clinical development.

**Clinical Trial Development, Job Rotation** 2024–2025
Novartis · Basel, CH

- Study start-up: protocol development, informed consent forms, pharmacy and laboratory manuals; exposure to regulatory and strategic decisions (endpoints, pooling of trials); organised executive committee meetings.
- Built a proof of concept combining automation, data science and generative AI to centralise trial data and automate data-quality checks (CRF cross-checks, extraction of key facts from SAE narratives).

**Senior Data Scientist, Biomarker Development** 2020–2025
Novartis Biomedical Research · Basel, CH

- Exploratory biomarker analyses (transcriptomics, proteomics) for Phase I–III trials in immunology, nephrology and cardiovascular disease, supporting regulatory, research and medical-affairs objectives.
- Results presented at EULAR 2023 (remibrutinib, Sjögren's), ASN Kidney Week 2023 (iptacopan, IgA nephropathy) and ACR 2023 (ianalumab, lupus and Sjögren's); contributed to the remibrutinib Phase 2 publication (*Ann Rheum Dis*, 2023).

**Deep Learning for Genomics; Bioinformatics Scientist** 2017–2020
Central European Institute of Technology (CEITEC) · Brno, CZ

- Machine-learning and deep-learning (CNN) models for genetic and transcriptomic data, including a multi-branch CNN to identify small non-coding RNA loci (*Scientific Reports*, 2020).

**Bioinformatician & IT Specialist**, University Hospital Fondazione Tettamanti · Milan, IT 2014–2020

**Review Editor, Computational Genomics**, Frontiers · Remote 2021–present

## EDUCATION

**PhD, co-joint degree** 2016–2020
Translational & Molecular Medicine, University of Milan-Bicocca (IT) · Genomics & Proteomics, Masaryk University (CZ). Thesis research: machine learning and deep learning for genomic data.

**M.Sc. Molecular Biology and Bioinformatics**, Università degli Studi di Milano

**B.Sc. Biotechnology**, University of Milan-Bicocca

## SKILLS

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| **Clinical development** | Phase I–III trials, cardiovascular outcome trials, study start-up, protocol execution, retention and endpoint collection, data quality, ICH guidelines and estimands |
| **Data & AI** | Machine learning, deep learning, generative AI and LLMs, retrieval-augmented generation, automation of data review; Python, R, Shiny, Quarto; omics and biomarker analysis |
| **Leadership** | AI product ownership, cross-functional alignment (data management, biostatistics, operations, regulatory, medical affairs), work with country teams and national leaders |

## SPEAKING & WRITING

Invited speaker, Faces of Industrial Research, Switzerland (2025). Essays on AI in clinical development, trial operations and the move from academia to industry on my personal website (andreagrioni.github.io), Medium and LinkedIn.